Probstat 9

Monday, 05 October 2026 11:39

Continuous random variable

- $x \in R$. only contains real numbers.
- Stated in interval / range.

Area:

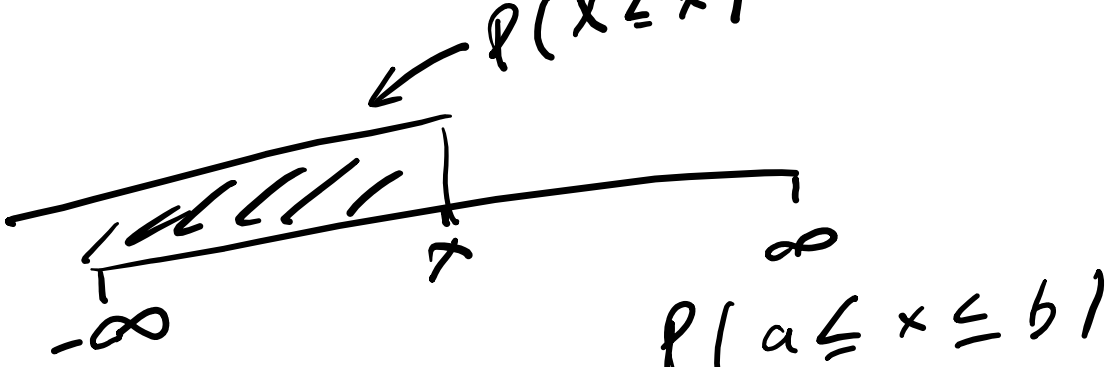
X: nilai uts: $0 \leq x \leq 100$

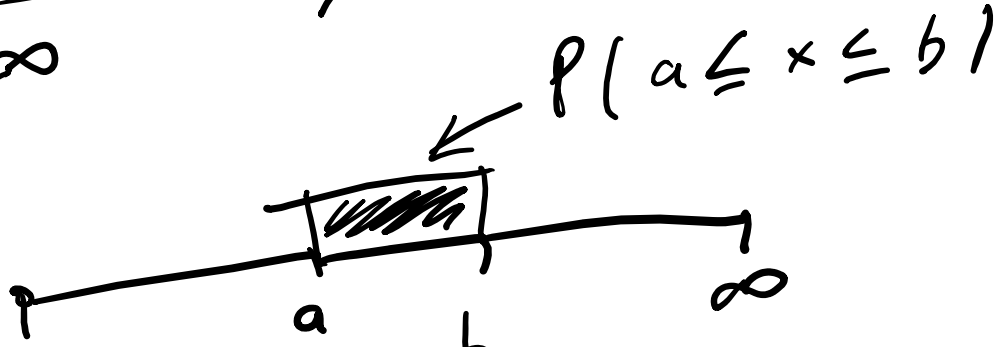
Y: Tinggi Badan: $140 \leq y \leq 200$

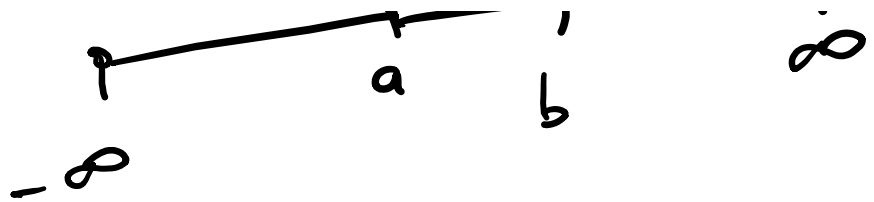
Z: Suhu ruang: $20 \leq z \leq 38$.

Fungsi Peluang Cont. Random. Var

$P(X \leq x) = P(x) = pdf$







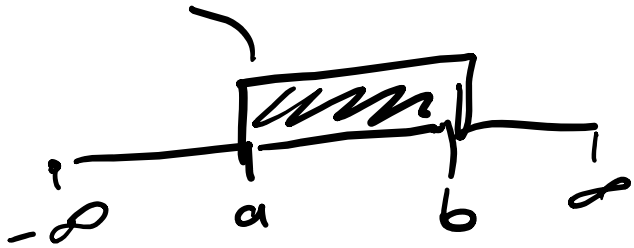
Fungsi kumulatif cont. random. var

$$F(a) = \int_{-\infty}^{x} f(x)\, dx$$



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$$F(x) \int_a^b f(x)\, dx$$



$$\int_{-\infty}^{\infty} f(x)\, dx = 1.$$

Mean & Variance Cont. rdm. var.

$$\mu = E(x) = \int_{-\infty}^{\infty} x \cdot f(x)\, dx$$

$$\sigma^2 = \mathrm{Var}(x) = \int_{-\infty}^{\infty} (x-\mu)^2 \cdot f(x)\, dx.$$

Expected values & variance

code di post